

GENERAL SPECIFICATION G3.22

Steel Reinforcement

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G3.22 STEEL REINFORCEMENT

G3.22.1 Scope

- (a) This section of the Specification covers detailing, fabricating, supplying and placing of reinforcing steel and accessories for all cast in place and precast concrete.
- (b) The Contractor shall be entirely responsible for the correctness of bar schedules before arranging for the supply, cutting and bending of steel reinforcement.

G3.22.2 Reference Standards

- (a) Unless otherwise specified, steel reinforcement incorporated in the works shall comply with the following standards:

Standard	Subject
BS 4449	Steel for the reinforcement of concrete – Weldable reinforcing steel. Bar, coil and decoiled product
BS 4483	Steel fabric for the reinforcement of concrete
BS 10080	Steel for the reinforcement of concrete – Weldable reinforcing steel
BS 7973-1	Spacers and chairs for steel reinforcement and their specification
BS 8666	Scheduling, dimensioning, bending, and cutting of steel reinforcement for concrete

G3.22.3 Submissions

- (a) The following submissions are required to be submitted by the Contractor.
 - (i) Test certificates
 - 1) Manufacturer's test certificate for each delivery of reinforcing steel. Each certificate shall show the country of origin and tests carried out; and
 - 2) Manufacturer's test certificate for mechanical couplers.
 - (ii) Data
 - 1) Manufacturer's data on accessories and mechanical couplings.
 - (iii) Details of proposed welding procedures and welder's qualifications.

G3.22.4 Steel Reinforcement

- (a) Steel for reinforcement shall be of the following kinds as may be specified or detailed on the Drawings.
- (b) Type H- (500A, 500B or 500C) either:
 - (i) Hot rolled deformed high yield steel bars (with geometrical classification Type 2); or
 - (ii) Cold rolled deformed high yield steel bars (with geometrical classification Type 2).
- (c) Fabric welded hard drawn steel wire and other cold worked high bond bar fabric.

G3.22.5 Accessories

- (a) The Contractor shall supply all accessories such as reinforcing steel supports, hold-downs, spreaders, hangers, tie wire and all other incidentals necessary to complete an acceptable installation of all steel reinforcement. All accessories shall be of steel unless otherwise specified.
- (b) Spacer blocks for maintaining concrete cover to reinforcement shall be of concrete of the same or greater strength, durability, porosity and appearance as the in-situ concrete. Concrete spacers shall be in the form of a truncated cone or pyramid and shall be used with the larger face towards the reinforcing steel. The smaller face of a truncated cone or pyramid shall have a maximum dimension of 50 mm.
- (c) Chairs and other accessories for maintaining reinforcement and pre-stressing tendons in position shall be of steel.
- (d) Binding wire shall be No 16 gauge (1.60 mm) soft iron wire.

G3.22.6 Detailing

- (a) Steel reinforcement shall be as shown on the Drawings. The Contractor shall be responsible for checking the Drawings before cutting and bending reinforcement. When any information relating to reinforcement is missing or apparently incorrect, the Contractor shall remedy the missing or incorrect information to the approval of the S.O..

G3.22.7 Cutting and Bending of Reinforcement

- (a) All reinforcement bars which required to be bent shall, in the first instance, be supplied in the correct shape from the manufacturer. Where bending, cutting or re-shaping on site is required, the Contractor shall obtain prior approval or further instruction from the S.O..
- (b) Bending shall be carried out slowly with the use of an approved bending machine at a steady even pressure, without jerk or impact. If necessary, reinforcement may be warmed to a temperature not exceeding 100 degrees Celsius to facilitate bending, after which it shall be allowed to cool slowly in air. Hot bars shall not be cooled by quenching with water. Rebending of bars shall not be permitted.
- (c) Under no circumstances shall the Contractor re-bend any reinforcement bar without prior approval or further instruction from the S.O.. Any bars showing signs of cracking or brittleness after bending or agreed re-bending shall be rejected. In the event that bending operations are carried out without the approval of the S.O., the Contractor shall be responsible for breaking out and making good any such concrete into which re-bent reinforcement has been cast.
- (d) Cutting of reinforcement bars shall be carried out by appropriately qualified personnel using a cropping machine fit for the purpose.
- (e) On Site facilities specifically designed for bending reinforcement shall be provided for minor adjustments.
- (f) Minor adjustments may be made to the projecting legs of continuity reinforcement by bending using Equipment defined in this Clause.
- (g) Cut and bent bars shall be bundled and labelled for positive identification with the Drawings and bending schedules, until they are incorporated into the Works.

G3.22.8 Storage of Reinforcing Bars and Fabric

- (a) The Contractor shall stack separately and label different types of reinforcement for positive identification.
- (b) Reinforcement shall not be subjected to mechanical damage or shock loading prior to embedment.
- (c) Steel reinforcing bars shall be kept clean and shall be free from pitting, loose rust, mill scale, oil, grease, earth, paint, or any other material which may impair the bond between the concrete and the reinforcement.
- (d) Reinforcing steel shall be stored and fabricated under cover on wooden or concrete supports such that the steel is elevated from the ground surface by a minimum of 150 mm.

G3.22.9 Fixing of Reinforcement

- (a) All reinforcement shall be securely and accurately fixed in position shown on the Drawings using approved spacer blocks or bar chairs. The spacer blocks or bar chairs shall be designed by the Contractor's Professional Engineer and submitted to S.O. for approval. The cost of providing these accessories are deemed to have been included in the steel reinforcement price, quoted by the Contractor in Bill of Quantities. All intersections of bars shall be secured with No. 16 gauge (1.60 mm) soft iron wire, the ends being turned into the body of the concrete. The Contractor shall ensure that all reinforcement is maintained in position at all times, particular care being taken during placing of concrete.
- (b) Concrete cover to reinforcement shall be as specified or detailed on the Drawings, and shall be maintained in accordance with the relevant tolerances. Correct concrete cover to reinforcement shall be maintained with the aid of approved spacer pieces. Top reinforcement in slabs shall be maintained in position by means of chairs, the diameter and quantity being sufficient to ensure security of the reinforcement in shape and position.
- (c) No part of the reinforcement shall be used to support formwork, access ways, working platforms, or the placing equipment or for the conducting of an electric current other than as part of an earthing system during the site welding of reinforcement.

G3.22.10 Mechanical Couplers for Reinforcement Bars

- (a) All mechanical couplers for tension and compression connections shall be of a type which are simple to install and which can be confirmed by quick visual inspection to have been correctly installed and to have achieved a connection providing the required full strength of the parent bars. Thread systems shall have parallel metric threads to facilitate fixing. Where couplers are required for remedial works, they shall be submitted for the S.O.'s acceptance, and shall in general comply with the requirements below.

- (b) The Contractor shall submit to the S.O. for his acceptance a method statement for mechanically connecting the reinforcement, and for the installation and verification inspection of couplers. This shall take into account any special requirements for horizontal, vertical and inclined couplers and shall include a rectification procedure if the connection is incorrectly made. It shall also cover the correct handling of tools and equipment for mechanical connecting on site. The following information shall also be included:
 - (i) For threaded couplers
 - 1) Requirements for cleanliness;
 - 2) Equipment for threading bars;
 - 3) Equipment and method for torqueing (as required), force required and method of measurement;
 - 4) Method of locking the connections on both rebars; and
 - 5) Method of verification of final rebar alignment and coupler integrity.
 - (ii) For swaged couplers
 - 1) Requirements for cleanliness;
 - 2) Bar end preparation and permissible deviation;
 - 3) Equipment and method of swaging, minimum and maximum number of swages per sleeve;
 - 4) Swaging pressure/stresses, method of measurement and tolerance on pressure/stress;
 - 5) Method of verification that the sleeve is swaged for the full length on both bars; and
 - 6) Method of verification of final rebar alignment and coupler integrity.
- (c) Couplers shall be installed strictly in accordance with the Manufacturer's recommendations.
- (d) Where couplers are cast-in to concrete, but the connection is not to be completed for a period of more than 28 days from the date of casting (or shorter time if appropriate), the couplers shall be internally greased and plastic capped to a protection detail acceptable to the S.O..
- (e) The Contractor shall ensure that the manufacturer of the couplers operates an ISO 9002 approved quality assurance program, or equivalent, for the manufacture of couplers. The Contractor shall ensure that the supplier/Installer operates an ISO 9002 approved quality assurance programme, or equivalent, for reinforcement bar end processing and for coupler installation.
- (f) All couplers shall be clearly stamped with the batch or heat number. This number shall be traceable to the original cast.
- (g) The relevant material mill certificates for couplers are to be submitted with each delivery. The certificates shall show the salient material properties of the couplers.

- (h) Packaging to couplers shall not be removed until they are to be used in the jointing process.
- (i) For threaded systems every prepared bar end shall undergo a load test prior to leaving the supplier's workshop. Minimum test loading shall be to 75% of the characteristic strength and maximum test loading shall be to 100% of characteristic strength. A positive indication shall be stamped on the rebar to indicate that this operation has been carried out.
- (j) Any special tools and equipment required for use with the couplers shall be in accordance with the recommendations of the Manufacturer for the accepted couplers. Such tools and equipment shall be serviced and calibrated at intervals recommended by the Manufacturer. The Contractor shall retain documentation in respect of servicing and calibration for inspection by the S.O..
- (k) The Contractor shall arrange for a suitably qualified Manufacturer's representative experienced in mechanically connecting reinforcement to be present at site before the start of work for initial training of personnel, and also to demonstrate the equipment and techniques as necessary.
- (l) Each coupler shall be visually examined prior to use to ensure the absence of rust and of any foreign material on the inside surface. All completed couplers shall be inspected and verified in accordance with the Contractor's accepted method statement. The Contractor shall establish to the satisfaction of the S.O. a procedure for documenting the inspection of the couplers and to show that every coupler installed complies with the requirements of this specification. The Contractor shall retain inspection records and shall submit copies to the S.O. within seven (7) days of the installation of the couplers. Couplers that do not meet the acceptance standards shall be completely removed and the bars re-connected as required.
- (m) All couplers shall satisfy the following criteria:
 - (i) Permanent Elongation Test
 - 1) When a test is made in accordance with the methodology in SS2 of a representative gauge length assembly (comprising reinforcement of the size, grade and profile to be used and a coupler of the precise type to be used), the permanent elongation after loading to 60% of the characteristic strength and returning the load to zero should not exceed 0.1 mm.
 - (ii) Static Tensile Test
 - 1) The tensile strength of the coupled bar in a representative gauge length assembly when tested to failure in accordance with the methodology in SS2 shall exceed 287.5 N/mm² for grade 250, 529 N/mm² for grade 460 hot rolled steel, and 506 N/mm² for grade 460 cold worked steel; in addition, the failure of the coupled bar assembly shall occur in the bar clear of the coupler and of the testing machine grips.

- (n) Permanent Elongation Tests and Static Tensile Tests shall be carried out as follows at the frequency specified:

- (i) Proving tests. Prior to use of any of the accepted systems on site, the supplier/Installer shall be required to prepare representative gauge length assemblies and subject each one to a Permanent Elongation Test and to a Static Tensile Test. In addition, the Supplier/ Installer shall be required to prepare control bars and to subject each one to a Static Tensile Test. For each type, size of bar and coupler type, the number of samples and tests required are:

	Representative gauge length assemblies		Control Bars
Min no. of samples per coupler type & size	2		1
Test Type	Static Tensile	Permanent Elongation	Static Tensile
Min no. of tests per sample	1	1	1

- (ii) Note: A control rebar refers to a specimen length of reinforcement bar cut from the same continuous length as that of the test specimen. All specimens and control bars are to be cut from 1 continuous length of reinforcing bar. Should it be necessary to use more than 1 length of bar, then a control bar shall be taken per length and tested for comparison against corresponding specimens.
- (iii) The Contractor shall submit to the S.O. for acceptance a report on the Proving Tests, within seven working days of the tests. The report shall provide full details including:
- 1) Results of all tests;
 - 2) Details of dimensions, geometry and torquing and if applicable swaging pressures;
 - 3) Details of test procedures;
 - 4) Description of test rig / load cell;
 - 5) Description of load monitoring, strain measurements; and
 - 6) Calibration certificates.
- (iv) From each test on the coupler and control bar specimens the following information shall be provided:
- 1) Stress – Strain (or Load Displacement) curves;
 - 2) Yield Load & Yield Strength;
 - 3) Elongation of the mechanical connection;
 - 4) Ultimate load & Ultimate Tensile Strength;

- 5) Mode of failure; and
 - 6) Gauge length used for strain measurement and statement of how gauge length was determined.
- (v) Production testing. Sacrificial couplers together with adequate lengths of reinforcement attached shall be installed next to working couplers at the same time and using the same equipment as the working couplers to form sample gauge length assemblies and control bars for production testing. The samples and the control bars shall be removed for production testing. Each sample assembly shall be subjected to a Permanent Elongation Test and to a Static Tensile Test, and each control bar to a Static Tensile Test. The sample frequency for each bar size shall be:
- 1) One gauge length assembly and one control bar for the first 10 couplers; and
 - 2) One gauge length assembly and one control bar for the next and subsequent units of 200 couplers.
- (vi) The Contractor shall submit to the S.O. for acceptance a monthly report of the production testing containing full details of the testing, including the same information as that for the proving tests above.

G3.22.11 Welding of Reinforcement

- (a) Welding on site shall be avoided wherever possible, but where suitable safeguards and techniques are employed and provided that the types of steel (including high-yield steels to SS 2) have the required welding properties, it may be undertaken with the acceptance of the S.O.. Before welding any reinforcement, the Contractor shall supply to the S.O. a welding procedure specification (WPS) and an example of the weld for the type of steel, connection and weld being proposed. If such evidence is not available, the Contractor shall demonstrate satisfactory performance by means of testing as accepted by the S.O..
- (b) In addition, and as required by the S.O., the competence of the operators shall be demonstrated prior to and periodically during welding operations by submission of independent welder qualification records (WQR) for each welder to be used on site.
- (c) Welding may be used in fixing reinforcement in position, for example, by welding between crossing or lapping reinforcement, or between bars and other steel members.
- (d) Structural welding shall not be carried out unless specifically shown on the Drawings.
- (e) Notwithstanding the above, the S.O. will not permit tack welding of bars which will be subject to fluctuating stresses in the completed structure.
- (f) Metal-arc welding of reinforcement shall be carried out in accordance with BS 5135 and the recommendations of the reinforcement manufacturer.
- (g) Other methods of welding may be used subject to the acceptance of the S.O..
- (h) Exposed reinforcement to be built into later concrete shall be effectively protected from corrosion, to the acceptance of the S.O..

G3.22.12 Inspection and Testing

- (a) When required by the S.O., the Contractor shall take samples from reinforcement delivered to site and shall arrange for the samples to be tested by an approved testing agency. Test certificates from that agency shall be submitted to the S.O..
- (b) Reinforcements shall be tested for Strength, Ductility, Weldability, Bendability and Bond Characteristics. For each diameter of reinforcement, 1 test shall be conducted for every 100 t or part thereof received at site, which shall be carried out for each source of supply. The test frequency shall be followed unless otherwise specified and agreed by S.O..
- (c) The S.O. or his authorised representative shall be required to witness such testing at SAC-SINGLAS accredited laboratory.
- (d) The Contractor shall be responsible for checking and inspection of reinforcement before requesting S.O. for inspection to allow placing of concrete.

END OF SECTION